

**ADVENTIST UNIVERSITY OF CENTRAL
AFRICA**

MASTERS IN BIG DATA ANALYTICS

**Comprehensive Solutions: Statistical Learning and
Linear Regression Exercises**

Programme:	MSc Big Data Analytics
Course Title:	Information Retrieval and Data Mining
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Question 1

For each case, we compare a **flexible** method (low bias, high variance) with an **inflexible** method (high bias, low variance), using the bias-variance trade-off.

(a) Extremely large n , small p

Expected winner: flexible method (generally better).

When n is very large, parameter estimates become stable even for models with higher effective complexity. Since p is small, the feature space is not highly sparse, so a flexible method can learn subtle structure without suffering severe variance inflation. In this regime, the variance penalty of flexibility is reduced by large sample size, while bias reduction remains valuable.

Conclusion: with enough data and few predictors, flexibility is usually advantageous.

(b) Extremely large p , small n

Expected winner: inflexible method (generally better).

Here we are in a high-dimensional, low-sample-size setting ($p \gg n$). Flexible methods can fit noise very easily, leading to overfitting and unstable estimates. The variance term becomes dominant, and prediction error typically increases out of sample. Inflexible or regularized models constrain complexity and are therefore more robust.

Conclusion: with many predictors and few observations, inflexible methods are usually preferable.

(c) Highly non-linear relationship between predictors and response

Expected winner: flexible method (generally better).

If the true signal is strongly non-linear, inflexible methods (for example strict linear forms) can incur large approximation bias. Flexible methods are designed to capture curvature, interactions, and local structure.

Conclusion: when non-linearity is strong and data quality is reasonable, flexible methods tend to outperform due to lower bias.

(d) Extremely high error variance $\sigma^2 = \text{Var}(\epsilon)$

Expected winner: inflexible method (generally better).

When irreducible noise is very high, aggressive flexibility often starts fitting random fluctuations rather than signal. This amplifies variance without reducing irreducible error. A simpler model is usually more stable and generalizes better.

Conclusion: high-noise settings generally favor less flexible approaches.

Question 2: Boston Housing Data

We work with the `Boston` data from ISLR2.

(a) Dimensions and meaning of rows/columns

- Number of rows: 506
- Number of columns: 13

Interpretation:

- Each row is a Boston census tract (observation/unit).
- Each column is a variable measured on that tract.
- Variables include socioeconomic, environmental, accessibility, and housing-value characteristics (for example `crim`, `rm`, `tax`, `lstat`, `medv`).

(b) Pairwise scatterplots of predictors and findings

A standard command is:

```
pairs(Boston)
```

Observed structure is not random; several strong pairwise trends appear. Correlation magnitudes supporting the visual patterns include:

Pair	Correlation	Interpretation
<code>rad</code> vs <code>tax</code>	0.910	very strong positive linear relation
<code>nox</code> vs <code>dis</code>	-0.769	strong negative relation
<code>indus</code> vs <code>nox</code>	0.764	strong positive relation
<code>age</code> vs <code>dis</code>	-0.748	older housing where distance to employment centers is lower
<code>lstat</code> vs <code>medv</code>	-0.738	socioeconomic disadvantage linked to lower house value
<code>rm</code> vs <code>medv</code>	0.695	more rooms linked to higher house value

Summary finding: the predictors display substantial multicollinearity structure and clear socioeconomic-geographic gradients.

(c) Are predictors associated with per-capita crime rate (`crim`)?

Yes. Several predictors are strongly associated with `crim`. Using pairwise correlations:

- Positive associations: `rad` (0.626), `tax` (0.583), `lstat` (0.456), `nox` (0.421), `indus` (0.407), `age` (0.353), `ptratio` (0.290).
- Negative associations: `medv` (-0.388), `dis` (-0.380), `rm` (-0.219), `zn` (-0.201).

- Near-zero: **chas** (-0.056), indicating weak linear relation with crime.

Simple one-predictor regressions confirm statistical significance for nearly all except **chas** (non-significant at conventional levels).

Interpretation: higher crime tends to occur in tracts with higher highway access index, higher tax burden, higher pollution/industrial activity, and lower housing quality/value indicators.

(d) High crime/tax/pupil-teacher ratio tracts and predictor ranges

Ranges (min to max):

Variable	Range
crim	0.00632 to 88.9762
zn	0 to 100
indus	0.46 to 27.74
chas	0 to 1
nox	0.385 to 0.871
rm	3.561 to 8.780
age	2.9 to 100
dis	1.1296 to 12.1265
rad	1 to 24
tax	187 to 711
ptratio	12.6 to 22.0
lstat	1.73 to 37.97
medv	5 to 50

Comments on extremes:

- Crime is extremely right-skewed: median ≈ 0.257 , max 88.98. So a small subset has exceptionally high crime.
- Tax also has a pronounced upper tail: median 330, upper quartile 666, max 711.
- Pupil-teacher ratio is narrower in spread: median 19.05, max 22.0.

So yes, some tracts are particularly extreme for crime and tax, while **ptratio** has less dramatic dispersion.

(e) Number of tracts bounding the Charles River

$$\sum_{i=1}^{506} \mathbf{1}(\text{chas}_i = 1) = 35.$$

So **35 tracts** bound the Charles River.

(f) Median pupil-teacher ratio

The median of `ptratio` is:

$$\text{median}(\text{ptratio}) = 19.05.$$

(g) Tract with lowest median home value (`medv`)

The minimum `medv` is 5.0, occurring at tract index 399 (in this row ordering), with:

Variable	Value
<code>crim</code>	38.3518
<code>zn</code>	0
<code>indus</code>	18.10
<code>chas</code>	0
<code>nox</code>	0.693
<code>rm</code>	5.453
<code>age</code>	100
<code>dis</code>	1.4896
<code>rad</code>	24
<code>tax</code>	666
<code>ptratio</code>	20.2
<code>lstat</code>	30.59
<code>medv</code>	5.0

Comparison to overall ranges:

- Very high crime, high pollution, high tax, high accessibility index, and very high lower-status percentage.
- Rooms per dwelling are low-to-moderate relative to the full range.
- Housing is old (`age`=100) and close to employment centers (low `dis`).

Finding: the lowest-value tract combines many characteristics typically associated with urban deprivation and lower property values.

(h) Tracts with more than seven rooms and more than eight rooms

$$\#\{i : \text{rm}_i > 7\} = 64, \quad \#\{i : \text{rm}_i > 8\} = 13.$$

Comment on `rm` ≥ 8 tracts:

- Most have high `medv` (often near the upper cap of 50), indicating premium housing stock.

- They frequently have low `lstat` (socioeconomic advantage).
- Many also have relatively low crime, with a notable exception: one tract has `rm`=8.780 but only `medv`=21.9, showing that high room count alone does not guarantee high value when other adverse factors are present.

Question 3 (Chapter 3, Exercise 1)

Table 3.4 in ISLR reports a multiple linear regression of `Sales` on `TV`, `radio`, and `newspaper`. Each p-value tests whether the corresponding predictor contributes linearly to explaining sales *after adjusting for the other predictors*.

Null hypotheses in words:

- For TV: given radio and newspaper budgets, TV has no linear association with sales.
- For radio: given TV and newspaper budgets, radio has no linear association with sales.
- For newspaper: given TV and radio budgets, newspaper has no linear association with sales.

From Table 3.4, TV and radio have extremely small p-values (highly significant), while newspaper has a very large p-value (not significant).

Practical conclusion (in business language):

- There is strong evidence that increasing TV spend is associated with higher sales, holding other media fixed.
- There is strong evidence that increasing radio spend is associated with higher sales, holding other media fixed.
- There is no convincing evidence that newspaper spend has an additional linear effect on sales once TV and radio are already in the model.

This does *not* mean newspaper advertising is universally useless; it means in this dataset and this linear specification, its incremental contribution beyond TV and radio is not statistically detectable.

Question 4 (Chapter 3, Exercise 6)

Show that the least-squares line in simple linear regression passes through $(\bar{x}, \bar{y})$.

Model:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i,$$

with least-squares estimates $\hat{\beta}_0, \hat{\beta}_1$ minimizing

$$RSS = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2.$$

The normal equation from differentiating with respect to β_0 is

$$\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0.$$

Rearrange:

$$\sum_{i=1}^n y_i - n\hat{\beta}_0 - \hat{\beta}_1 \sum_{i=1}^n x_i = 0.$$

Divide by n :

$$\bar{y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{x} = 0 \quad \Rightarrow \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Now evaluate the fitted line at $x = \bar{x}$:

$$\hat{y}(\bar{x}) = \hat{\beta}_0 + \hat{\beta}_1 \bar{x} = (\bar{y} - \hat{\beta}_1 \bar{x}) + \hat{\beta}_1 \bar{x} = \bar{y}.$$

Hence the least-squares line always passes through $(\bar{x}, \bar{y})$.

Question 5 (Chapter 3, Exercise 11)

Given:

```
set.seed(1)
x <- rnorm(100)
y <- 2*x + rnorm(100)
```

(a) Regress y on x without intercept

Model fitted: `lm(y ~ x + 0)`.

Estimated coefficient table:

Term	Estimate	Std. Error	t value	p-value
x	1.993876	0.106477	18.72593	2.64×10^{-34}

Comment: The slope estimate is very close to the data-generating value 2. The t-statistic is extremely large in magnitude and the p-value is tiny, so we strongly reject $H_0 : \beta = 0$.

(b) Regress x on y without intercept

Model fitted: `lm(x ~ y + 0)`.

Estimated coefficient table:

Term	Estimate	Std. Error	t value	p-value
y	0.3911145	0.0208863	18.72593	2.64×10^{-34}

Comment: Coefficient scale differs because the response/predictor roles changed, but inferential strength against the null is identical (same t and p).

(c) Relationship between (a) and (b)

The slope estimates are not equal and generally not exact reciprocals in noisy finite samples, but the t-statistics and p-values for testing zero slope are the same in both no-intercept regressions.

(d) Algebraic form of the no-intercept t-statistic

For regression of Y on X without intercept,

$$\hat{\beta} = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2},$$

$$SE(\hat{\beta}) = \sqrt{\frac{\sum_{i=1}^n (y_i - x_i \hat{\beta})^2}{(n-1) \sum_{i=1}^n x_i^2}}.$$

Hence

$$t = \frac{\hat{\beta}}{SE(\hat{\beta})}.$$

Compute the residual sum of squares:

$$\begin{aligned} \sum_{i=1}^n (y_i - x_i \hat{\beta})^2 &= \sum y_i^2 - 2\hat{\beta} \sum x_i y_i + \hat{\beta}^2 \sum x_i^2 \\ &= \sum y_i^2 - \frac{(\sum x_i y_i)^2}{\sum x_i^2}. \end{aligned}$$

Therefore,

$$SE(\hat{\beta}) = \sqrt{\frac{\sum y_i^2 - \frac{(\sum x_i y_i)^2}{\sum x_i^2}}{(n-1) \sum x_i^2}} = \sqrt{\frac{(\sum x_i^2)(\sum y_i^2) - (\sum x_i y_i)^2}{(n-1)(\sum x_i^2)^2}}.$$

Now divide:

$$t = \frac{\frac{\sum x_i y_i}{\sum x_i^2}}{\sqrt{\frac{(\sum x_i^2)(\sum y_i^2) - (\sum x_i y_i)^2}{(n-1)(\sum x_i^2)^2}}}$$

$$= \frac{\sqrt{n-1} \sum x_i y_i}{\sqrt{(\sum x_i^2)(\sum y_i^2) - (\sum x_i y_i)^2}}.$$

This is exactly the requested expression.

Numerical confirmation in R:

$$t_{\hat{\beta}/SE(\hat{\beta})} = 18.72593, \quad t_{\text{closed form}} = 18.72593.$$

(e) Why t is the same for y on x and x on y (no intercept)

From the derived formula,

$$t = \frac{\sqrt{n-1} \sum x_i y_i}{\sqrt{(\sum x_i^2)(\sum y_i^2) - (\sum x_i y_i)^2}}.$$

This expression is symmetric in x and y because multiplication is commutative and the denominator uses the same symmetric products. Swapping x and y leaves t unchanged.

Hence both no-intercept regressions have identical t-statistics (and therefore p-values) for testing zero slope.

(f) With intercept, show t-statistics also match

Using:

```
summary(lm(y ~ x))$coef["x","t value"]
summary(lm(x ~ y))$coef["y","t value"]
```

we obtain:

$$t_{y \sim x} = 18.5556, \quad t_{x \sim y} = 18.5556.$$

So with intercept included, the slope t-statistics are again identical. Conceptually, both tests are equivalent to testing zero correlation between x and y .

Closing Note

The results across Questions 2–5 are coherent with core principles in statistical learning: bias-variance trade-off, multicollinearity in real-world predictors, interpretation of conditional significance in multiple regression, geometry of least squares, and symmetry properties of t-tests under role reversal of two variables.